A Sufficient Statistic for Algorithmic Feedback: What Prices Reveal About Trader Groups, and Who Needs the Rest

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Working paper, core version: 8 November 2026

Abstract

Algorithmic traders react to prices, and prices react to them. In a linear market with feedback traders, one object, the *feedback kernel* (the response of total feedback flow to past prices), is a sufficient statistic for everything that price-based trading determines. For any insider strategy and any causal dealer rule, two trader populations with the same kernel produce the same prices, order flow, insider profits and dealer inventories, and therefore the same equilibria. The kernel is also all that prices identify. Groups with equal memories cannot be separated with any amount of data. Separating K groups whose memories lie $\Delta\eta$ apart takes of order $(\Delta\eta)^{-(4K-2)}$ experiments. That is the known rate for finite mixtures, and it survives the feedback loop because the memories enter as zeros of the price response; for the same reason, watching the market is no cheaper than trading to learn. In an equilibrium with inventory-averse dealers, the price is the same closed loop with effective parameters and unchanged memories, so both results hold in equilibrium, also with a strategic insider whose information is short-lived. Composition therefore matters only for actions aimed at a particular group: two markets almost identical in their prices can call for different halts. Flow reported by trader class removes the cost of learning it. The idea of a sufficient statistic for aggregate dynamics is not new. It parallels intertemporal marginal propensities to consume in macroeconomics and the aggregation of autoregressions in time series. Our

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contribution is its strategic, closed-loop and supervisory form. The core statements are machine-checked in Lean 4; the supplement lists what is proved only on paper.

AI-use disclosure. Claude, ChatGPT/Codex and Gemini provided material assistance with research development, code, numerical checks, manuscript drafting and internal critique, including the restructuring of this core version. ChatGPT/Codex additionally developed and checked Lean proofs. Human authors remain responsible for verifying the content and its scholarly contribution before submission. AI checks do not constitute independent review, and no independent human proof check has been completed.

Relation to the full working paper. This is a shortened version, organized around one result. Every result stated here also appears in the full working paper ("What Prices Cannot Tell a Market Supervisor", 5 November 2026), with the same proof. The one exception is Proposition 3.2, which is new to this version. Results not needed for the main argument are omitted, among them account-level capital, general memory, spectral blindness and every empirical test. Section 7 says where they are.

Keywords: algorithmic trading, feedback trading, sufficient statistics, aggregation, identification, market microstructure, supervision, disclosure.

JEL: C32, D83, G14, G18.

Code and proofs. All code, Lean proofs and result files are public at https://github.com/Pratikn03/Fianince.

1 Introduction

Much of today's trading is done by programs that read recent prices, decide what to trade, and by trading move the prices they read. Trend-following funds, execution algorithms, market-making programs and, more recently, AI trading agents all work this way. When such a market misbehaves, supervisors ask who was trading and how fast those traders react. This paper asks a prior question: what does a market's price record reveal about its population of feedback traders, and for which decisions is that enough?

The answer in one sentence. Prices reveal the market's *feedback kernel*: the response, m periods later, of total feedback flow to a unit move in the price signal. That kernel is all that prices identify, and it is all that anyone trading on public information needs. Group composition, meaning who the groups are and how strongly each trades, matters only for actions aimed at a particular group, and learning it from prices has a known and steep price.

The model. The market has a price and K groups of traders. Each group keeps an exponentially weighted average of past prices, with memory rate r_i, and trades in proportion to it with weight w_i. A desk that tracks a trend with a Kalman filter, or that learns by regularized online gradient descent, behaves exactly this way (Lemma 2.1). Total feedback flow is then a convolution of past prices with the kernel

$$k_m = \sum_i w_i r_i^m, \qquad m = 0, 1, 2, \ldots,$$

the moments of the *merged measure* M, which puts weight w_i at rate r_i. The paper rests on three results about this one object.

(1) *The kernel is sufficient* (Theorem 3.1 and Proposition 3.2). Add a value, noise traders, an insider and a dealer, each following an arbitrary causal rule, which covers Kyle, inventory-averse, learning and misspecified dealers. Two trader populations with the same kernel then produce the same price and order-flow paths, the same insider trades and profits and the same dealer inventories, realization by realization. The two games have the same strategy sets and payoffs, and hence the same equilibria. The proof needs only that aggregate feedback flow is a fixed linear function of past public signals, so the result holds for any linear trader population, not only exponential memories.

(2) *The kernel is all that prices identify* (Theorem 3.3). Two populations give identical price responses to every trade sequence if and only if their merged measures coincide.

Groups with the same memory cannot be separated however long the record. Groups with different memories can be separated in principle.

(3) *Learning more has a known price* (Theorems 5.1, 5.2 and 5.4). To separate K groups whose memories lie within $\Delta\eta$ of each other takes of order $(\Delta\eta)^{-(4K-2)}$ experiments, and Prony's method attains this order. Waiting instead of trading does not help: passive observation needs at least the same order of observations.

The results survive equilibrium. In the base model the price impact and the persistence of mispricing are primitives. Section 4 derives both from competitive dealers who dislike holding inventory. The equilibrium price is then the same closed loop, path by path, with effective impact and persistence and with the memory rates unchanged (Theorem 4.2). So the kernel remains the sufficient statistic, and the cost of going beyond it keeps its order. The supplement adds a strategic insider with short-lived information. It also shows that, with positive feedback, liquidity disappears beyond a threshold in dealer risk aversion that falls with the feedback intensity.

Supervision splits in two. A market-wide action, such as a uniform cut in activity to meet a ceiling on the feedback budget, depends on the population only through the kernel. A targeted action, such as halting the largest group, does not. Two populations whose price responses differ only by order $(\Delta\eta)^3$ can need different halts (Proposition 6.1). For such decisions, flow reported separately by trader class turns each group into an ordinary regression, with no penalty as the groups' memories approach each other (Proposition 6.2). It cuts the number of experiments needed by a factor of order $(\Delta\eta)^{-(4K-2)}$ (Theorem 6.3).

1.1 What is known and what is new

The central idea of this paper, that an aggregate is determined by a distribution of individual coefficients and not by the individuals, is old. We state its precedents first, because they fix what we can claim.

Aggregation of dynamic models. Granger (1980) showed that aggregating many first-order autoregressions with different coefficients produces a process whose dynamics are governed by the distribution of the coefficients; with enough mass near one the aggregate has long memory. Zaffaroni (2004) develops the theory for large economies. Recovering the coefficient distribution from aggregate data is known as the disaggregation problem (Celov

et al., 2008). Our merged measure M plays the role of that coefficient distribution, and Theorem 3.3(i) is the corresponding identification statement for a closed feedback loop.

Sufficient statistics for aggregate dynamics. In macroeconomics, Auclert et al. (2024) show that the matrix of intertemporal marginal propensities to consume is a sufficient statistic for the output response to fiscal policy, whatever the household heterogeneity behind it. The kernel k plays the same role for a market with feedback traders. It is the response of aggregate demand to past prices, and it summarizes every heterogeneity that matters for prices.

Mixture estimation. The order $(\Delta\eta)^{-(4K-2)}$ is the known minimax rate for finite mixtures with K near-colliding components (Heinrich and Kahn, 2018; Wu and Yang, 2020), and super-resolution theory gives the same conditioning for clustered exponentials (Moitra, 2015; Batenkov et al., 2021; Katz et al., 2023). We do not claim the exponent.

Feedback trading and insiders. Positive-feedback traders and their effect on prices go back to De Long et al. (1990). Barberis et al. (2015, 2018) model extrapolators with exponentially weighted memories, and Brock and Hommes (1998) model heterogeneous rules. The insiders and dealers covered by Section 3 include those of Kyle (1985) and of inventory models in the tradition of Ho and Stoll (1981) and Grossman and Miller (1988).

What is new. Four things, each of which uses the market setting.

(a) *Sufficiency inside a strategic market.* The aggregation literature concerns the statistical law of an aggregate series. Theorem 3.1 is a statement about games: it holds path by path for every insider strategy and every causal dealer rule, so it transfers to every equilibrium concept defined from payoffs and public histories. Part (iv) shows where it stops, since group-level quantities are not invariant.

(b) *Memories are zeros of the closed loop.* In mixture and open-loop problems the unknown rates are locations or poles of the observed law. In a feedback market they are transmission zeros of the price response (Theorem 5.3). That decides which estimators are valid: fitting poles to prices finds the closed-loop eigenvalues and misses the memories. Because the closed loop is minimum phase, passive data face at least the same barrier as active probing.

(c) *The results hold in equilibrium.* With inventory-averse dealers who price the feedback traders' predictable flow, the equilibrium is still the closed loop, with unchanged memories (Theorem 4.2).

(d) *A supervisory split with a price.* The kernel suffices for market-wide actions and not for targeted ones, and class-tagged disclosure removes the cost of the targeted information (Propositions 6.1 and 6.2, Theorem 6.3).

The induction behind Theorem 3.1(iii) is short. Its content is not the induction but the object it isolates. The kernel is identified from prices, it is sufficient for every price-based outcome, and it is insufficient exactly for the targeted decisions supervisors care about.

Machine checking. The core statements are checked in Lean 4 against Mathlib with standard axioms only. The checks cover the sufficiency theorem (pathwise, for arbitrary causal rules), the K-group lower and upper bounds, the minimum-phase identification, the finite-sample passive bound, the reduced form of the inventory equilibrium, and the reduced form and best response in the insider equilibrium. Each statement says what is checked and what is proved only on paper; Supplement S6 lists the declarations.

Outline. Section 2 sets out the model. Section 3 proves that the kernel is sufficient and that it is what prices identify. Section 4 shows that both hold in equilibrium. Section 5 prices the information beyond the kernel. Section 6 shows which supervisory decisions need it and what disclosure buys. Section 7 states the limits, and Section 8 concludes.

2 Model and motivating examples

The model in words. The market has one price and K groups of traders. Each period three things happen. First, the price moves: it drifts back toward its fundamental value, and it is pushed by the groups' trades, by any test trade and by news. Second, each group updates its view by mixing the newest price into its running average, with weight η_i on the newest price and $1-\eta_i$ on its old view. Third, each group trades in proportion to its view, with intensity q_i. A group with a large η_i has a short memory and reacts quickly, like a short-horizon trend follower; a group with a small η_i has a long memory and reacts slowly. The equations below write this down.

Time is discrete. Let p_t be the price, p^* the fundamental value and $x_t = p_t - p^*$ the pricing error. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, an imposed primitive rather than an equilibrium quote, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{1}$$

where v_t is an executed probe quantity and w_t a price innovation. Cohort i holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1-\eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0,1). \tag{2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1-\eta_i$ for the memory rate and $w_i = q_i\eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j\geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z-a)D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \kappa/(1-S) = \sum_j h_j$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = Jz_{t-1} + bv_t$, where, because cohorts update on the same period's price, the probe enters every state on impact through

$$b = \kappa\,(1, \eta_1, \ldots, \eta_K)^\top, \tag{5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1-\eta_1+\eta_1\kappa q_1 & \cdots & \eta_1\kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K\kappa q_1 & \cdots & 1-\eta_K+\eta_K\kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{6}$$

In the *positive economy* ($q_i \geq 0$), J is entrywise nonnegative, its price row sums to S, and the economy is Schur stable iff $S < 1$.

Lemma 2.1 (Two motivating examples: filtering and learning)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0,1). \tag{7}$$

(ii) Gradient memory. *Online gradient descent with L_2 regularization $\gamma \in (0,1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and*

$q = \alpha H/\eta$. A tabular value update with learning rate η at a fixed state has the same form; general reinforcement-learning policies need not.

Why exponential memory, and what is assumed. Lemma 2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same reduced-form state. In these two families the memory rate is a known function of a tuning parameter, the desk's signal-to-noise ratio or its regularization. That is the private information a supervisor would like to recover, although real algorithms need not belong to either family. Applied to the pricing error, a desk that cannot separate fundamental news from transitory deviations treats recent deviations as a persistent trend, which is the exponential extrapolation of Barberis et al. (2015, 2018). Nothing in the model says such trading is profitable.

The results use six modelling choices: linearity; price-first timing within a period; exponential memory (needed for the identification result, Theorem 3.3, and the cost results of Section 5, but not for sufficiency); algorithms that do not change their rules when observed; known primitives (a, κ); and stated observation contracts. Unknown primitives can only enlarge the set of candidate populations, so the lower bounds remain valid; Corollary S1.2 in the supplement shows that estimating them leaves the upper-bound rate unchanged. Section 4 replaces the primitive (a, κ) by an equilibrium.

The observer. The cost results are stated for one observer outside the trader groups: a supervisor, an exchange's surveillance unit or a large desk. It knows (a, κ), the law of the observation noise and, for the K-group results, K. It can *wait*, recording prices driven by natural innovations; *probe*, trading a sequence v_t against the groups from a reset state and recording the response; or *acquire disclosure*, receiving each class's flow. Independent resets are a benchmark that no outside participant can run in a live market. We use them because they make the information content of the data exact.

3 The kernel is a sufficient statistic

This section states the paper's main result. In a market with an insider, noise traders, a dealer and feedback traders, all following arbitrary causal rules, the feedback traders affect every outcome only through their kernel. The notation for the insider follows Kyle (1985), so in this section x_t is the insider's order, not the pricing error.

Theorem 3.1 (The kernel is sufficient)**.** *Consider a market with the following participants and rules.*

- *A value v and noise trades z_t.*
- *An insider who submits $x_t = X_t(v, \mathcal{H}_{t-1})$.*
- *A dealer who sets $p_t = P_t(\mathcal{H}_{t-1}, y_t)$ from the public history $\mathcal{H}_{t-1}$ and the total order flow $y_t = x_t + z_t + f_t$.*
- *Feedback cohorts whose memory states update on a public price statistic $s_t = S_t(p_0, \dots, p_t)$ by $m_{i,t+1} = r_i m_{i,t} + s_t$ from $m_{i,0} = 0$, with $r_i = 1 - \eta_i$ and aggregate flow $f_t = \sum_i w_i m_{i,t}$, $w_i = q_i \eta_i$.*

The rules X, P and S are arbitrary causal maps. This covers Kyle, inventory-averse, learning and misspecified dealers, and any insider strategy.

(i) $f_t = \sum_{m<t} k_m \, s_{t-1-m}$, *with* $k_m = \sum_i w_i r_i^m$, *the moments of the merged measure M.*

(ii) Two cohort configurations produce the same aggregate flow on dates $t \le T$ for every signal sequence (s_t) if and only if $k_m = k'_m$ for all $m < T$. On an unbounded horizon this holds if and only if their merged measures coincide. For a fixed signal rule S, the 'if' direction holds for every price path. The 'only if' direction needs a rule that can produce an impulse signal. It fails, for example, for $S \equiv 0$, under which every configuration has zero flow.

(iii) If $k_m = k'_m$ for $m < T$, then for every strategy profile and every realization of (v, z) the following coincide on $t \le T$: the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory. The two games therefore have the same strategy sets and the same payoff for every profile. They have the same equilibria, with the same outcomes, under any equilibrium concept defined from these payoffs and the public history.

(iv) Group-level quantities are not invariant. A Gauss and a Radau two-point configuration agree on k_0, k_1, k_2, so they are outcome-equivalent for $T = 3$, while their largest cohorts carry weights $\frac{1}{2}$ and $\frac{3}{4}$.

In words. Whatever the insider and the dealer do, two markets whose trader groups have the same merged response produce the same prices, the same order flow and the same insider and dealer outcomes. So the part of the population that prices cannot identify (Theorem 3.3) is worth nothing to anyone who trades on public information. It matters only for actions aimed at a particular group, such as halting it (Proposition 6.1). Parts (i)–(iv) are machine-checked pathwise for every causal insider, dealer and signal rule (`Paper1EquilibriumInvariance.lean`). Supplement S2 gives the proof, the unbounded-

horizon statement, the extension to cohort-specific demand shocks and a numerical illustration.

Beyond exponential memory. The proof of part (iii) uses only one property of the feedback traders: their aggregate flow is a fixed linear function of past public signals. So the result does not depend on exponential memory.

Proposition 3.2 (Any linear population)**.** *In the market of Theorem 3.1, replace the cohorts by any finite population of feedback traders $j = 1, \dots, J$ whose flows are causal linear functions of the public signal,*

$$f_{j,t} = \sum_{m<t} \kappa_m^{(j)} \, s_{t-1-m}, \qquad \kappa^{(j)} = (\kappa_0^{(j)}, \kappa_1^{(j)}, \dots) \text{ arbitrary real sequences,}$$

and write $k = \sum_j \kappa^{(j)}$ for the population's kernel.

(i) Two populations produce the same aggregate flow on dates $t \le T$ for every signal sequence if and only if their kernels agree for $m < T$.

(ii) If their kernels agree for $m < T$, then for every strategy profile and every realization the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory coincide on $t \le T$, and the two games have the same equilibria under any equilibrium concept defined from payoffs and the public history.

(iii) Any decomposition of a kernel into individual kernels is consistent with the same outcomes. Prices, flows and all equilibrium outcomes therefore carry no information about the decomposition beyond k.

Proof. (i) Aggregate flow is $f_t = \sum_{m<t} k_m s_{t-1-m}$. If the kernels agree for $m < T$, the flows agree on $t \le T$ for every signal. Conversely, the impulse signal $s_j = \mathbf{1}\{j = 0\}$ gives $f_{m+1} = k_m$. (ii) The strong induction in the proof of Theorem 3.1(iii) (Supplement S2) uses only that f_t is the same function of the signals before t in both markets, which is part (i). (iii) follows from (ii), since outcomes depend on the population only through k. □

What the exponential case adds. For general linear traders, nothing about composition beyond k can be learned from anything that prices determine, because any split of k is observationally equivalent. Exponential memory is the case in which more can be learned, and the rest of the paper prices it. Then $k_m = \int r^m \, dM(r)$ is a moment sequence, and the kernel determines the merged measure M (Theorem 3.1(ii) on an unbounded horizon). A finite record determines only finitely many moments, which is why separating close groups is

costly. Proposition 3.2 is proved on paper. The machine-checked statement is Theorem 3.1, for exponential cohorts.

The analogy with aggregate demand. The kernel is the response of aggregate feedback demand to past prices. Auclert et al. (2024) show that the analogous object for households, the matrix of intertemporal marginal propensities to consume, is a sufficient statistic for the transmission of fiscal policy. In the same way, k summarizes every heterogeneity among feedback traders that matters for prices. The merged measure is the distribution of individual autoregressive coefficients that governs an aggregate in Granger (1980). Two points distinguish the market version. Sufficiency holds inside a game with an insider and a dealer, not only for the law of an aggregate series. And as part (iv) of Theorem 3.1 shows, it fails for group-level quantities, which matter for targeted supervision (Section 6).

3.1 What prices identify

Theorem 3.1 says that populations with the same kernel cannot be told apart by anything that price-based trading determines. The converse is that prices determine the kernel. In the probing experiment of Section 2, the response of prices to trades is a fixed function of (a, κ, k), and it determines the merged measure.

Theorem 3.3 (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state x_t.*

(i) Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1]. \tag{8}$$

Write $a' = a + \kappa M(\{0\})$, so that cohorts with $\eta = 1$ (rate 0) act as part of the anchor. For $a' \neq 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M) \cap (0, 1)|$, counting distinct positive rates with nonzero merged weight after signed cancellation; for $a' = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M) \cap (0, 1)|$. A cohort with $\eta = 1$ merges into the anchor a without adding a pole.

(ii) Separation complexity under average power. *Consider testing a single cohort with rate $\bar{\eta}$ against two cohorts with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with all rates in $[\epsilon, 1 - \epsilon]$ for some $\epsilon > 0$. Observe*

$\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \le P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs

$$T \;\ge\; \frac{4\sigma_\nu^2(1-2\alpha)^2}{C_{\mathcal{F}}^2 P\,\delta^4} = \Omega(\delta^{-4}), \tag{9}$$

where $C_{\mathcal{F}}$ depends only on the family, through ϵ, the bound on $|H|$ implied by uniform stability, and the total weight q.

In words. If two trading desks update on prices with the same memory, no amount of price data can tell them apart or say how much each trades: prices reveal only how much trading happens at each memory speed. Desks with different memories can be told apart in principle, and the rest of the paper prices how hard that is.

The memory rates are poles of G and *zeros* of H, not closed-loop poles. This has a practical consequence verified in the full working paper: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

Remark 3.4 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The δ^{-4} statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

4 Sufficiency and identification in equilibrium

The identification results of Section 3 take the price impact κ and the persistence a of pricing errors as given. This section derives both from dealers who dislike holding inventory. It shows that the equilibrium price is the closed loop of Section 2 with effective parameters and unchanged memory rates. So the kernel stays the sufficient statistic in equilibrium, and the cost of going beyond it (Section 5) keeps its order.

An inventory equilibrium.

- *Fundamental and price.* The fundamental F_t is a public random walk with innovation variance σ_F^2, and the price is $p_t = F_t + x_t$.

- *Cohorts.* They trade $u_{i,t+1} = r_i u_{i,t} + w_i x_t$, the paper's exponential memory with $w_i = q_i \eta_i$ and $r_i = 1 - \eta_i$.
- *Exogenous flow.* $\xi_t \sim N(0, \sigma_\xi^2)$, i.i.d., stands for noise trades or the observer's probe. Customer flow is $y_t = \sum_i u_{i,t} + \xi_t$.
- *Dealers.* Competitive dealers absorb the flow and lay off a share $1-\bar{a}$ of their inventory each period, so $H_t = \bar{a} H_{t-1} - y_t$. Each overlapping generation of dealers has CARA risk aversion A and holds H_t to the next date: $H_t = (E_t p_{t+1} - p_t)/(AV)$, with $V = \mathrm{Var}_t(p_{t+1})$.

We study stationary equilibria in which x_t is linear in the dealers' inventory and the cohorts' next trades. Write $G(D) = \sum_i w_i/(D - r_i) = \int dM(r)/(D - r)$ for the Stieltjes transform of the merged measure.

Theorem 4.1 (Price impact and persistence in an inventory equilibrium)**.** *Consider stationary linear equilibria of the model above.*

(i) Characterization. *The price law*

$$x_t = -\kappa H_t + \kappa \sum_i \frac{u_{i,t+1}}{D - r_i}$$

with $D \neq 0$ and $D \neq r_i$, is an equilibrium exactly when the induced dynamics are stable and

$$D = 1 - \kappa\, G(D), \qquad \kappa(D - \bar{a}) = A\, V\, D, \qquad V = \sigma_F^2 + \frac{\kappa^2 \sigma_\xi^2}{D^2}. \tag{10}$$

An unexpected unit of flow moves the price by κ/D.

(ii) No feedback. *Then $D = 1$ and the price law is exactly the paper's $x_t = \bar{a}\, x_{t-1} + \kappa\, y_t$. The persistence of pricing errors is the dealers' carry share, $a = \bar{a}$, and the impact κ solves $\kappa(1-\bar{a}) = A(\sigma_F^2 + \kappa^2 \sigma_\xi^2)$. An equilibrium exists if and only if $\Delta = (1-\bar{a})^2 - 4A^2\sigma_F^2\sigma_\xi^2 \geq 0$. When $\Delta > 0$ there are two, $\kappa_\pm = \big((1-\bar{a}) \pm \sqrt{\Delta}\big)/(2A\sigma_\xi^2)$; the low-impact one $\kappa_- \approx A\sigma_F^2/(1-\bar{a})$ is the price of dealer inventory risk.*

(iii) Invariance. *The coefficients (κ, D, V), and hence the price process, depend on the cohorts only through G, and so only through the merged measure M. This is Theorem 3.1 in equilibrium form.*

(iv) Small feedback. *Let $\Delta > 0$, fix $\delta > 0$ with $\max_i r_i < 1 - \delta$, and let $\epsilon = \sup_{|D-1| \leq \delta}(|G(D)| + |G'(D)|)$ be small. Then there is a unique equilibrium near*

$(\kappa_-, 1)$. *With* $G(1) = \sum_i q_i$, *the aggregate feedback intensity,*

$$D = 1 - \kappa_- G(1) + o(\epsilon), \qquad \kappa = \kappa_- + \frac{\kappa_-^2(\bar{a} + 2A\kappa_-\sigma_\xi^2)}{\sqrt{\Delta}}\, G(1) + o(\epsilon).$$

Positive feedback raises the impact κ, *and it raises the impact of an unexpected flow shock,* κ/D, *by more.*

In words. Dealers set the price impact and the persistence of mispricing. Impact is what they charge for carrying inventory risk; persistence is how slowly they lay that inventory off. Without feedback traders, this equilibrium is exactly the closed loop the paper starts from. Feedback traders enter only through their merged measure, as Theorem 3.1 requires. Positive feedback raises price impact, and dealers price their predictable future trades in advance (the terms in $u_{i,t+1}$). Feedback trading also makes liquidity fragile, not only more expensive: beyond a threshold in dealer risk aversion that falls with the feedback intensity, no equilibrium exists (Theorem S2.2).

Machine check. `Paper1InventoryEquilibrium.lean` checks the algebraic core of part (i):

- the dealers' first-order condition holds in every state;
- the next price loads on the next flow shock with coefficient κ/D;
- the no-feedback existence condition.

Part (iii) is proved on paper. For part (iv), `Paper1InventoryIFT.lean` applies Mathlib's inverse function theorem along a one-parameter family of feedback $G = \epsilon g$, with g strictly differentiable at $D = 1$. For all small ϵ it gives a zero near $(\kappa_-, 1)$, its uniqueness in a fixed neighbourhood, and the two first-order derivatives in the statement (`p1_inventory_small_feedback`). The version that is uniform over all small cohort measures, in the norm of the statement, is proved on paper.

The identification results hold in equilibrium. With feedback, dealers price the cohorts' predictable trades in advance, so the response of prices to flow is no longer the reduced form of Section 2 with the primitive (a, κ). The next theorem shows that it is still that closed loop, exactly and path by path, with new parameters and the same memory rates.

Theorem 4.2 (The equilibrium is the paper's closed loop)**.** *Take an equilibrium of Theo-*

rem 4.1 with $D \neq \bar{a}$, and set

$$a^* = \frac{\bar{a}}{D}, \qquad \kappa^* = \frac{\kappa}{D}, \qquad \tilde{w}_i = \frac{D - \bar{a}}{D - r_i}\, w_i.$$

(i) Reduced form, pathwise. *Let $\tilde{u}_{i,t} = \frac{D-\bar{a}}{D-r_i} u_{i,t}$. For every realization, every initial state and every t,*

$$x_t = a^* x_{t-1} + \kappa^* \Big(\xi_t + \sum_i \tilde{u}_{i,t} \Big), \qquad \tilde{u}_{i,t+1} = r_i \tilde{u}_{i,t} + \tilde{w}_i x_t.$$

So the price follows the closed loop (1)–(2) with persistence a^*, impact κ^*, exogenous flow ξ in place of the probe, and cohorts with the same memory rates and merged weights $\tilde{w}_i$. Its transfer function from flow to price is (4) with $(a^*, \kappa^*, \tilde{w})$. The equilibrium is stable if and only if this closed loop is.

(ii) Zeros. *For $\bar{a} > 0$, distinct rates in $(0, 1)$ and nonzero weights, the memory rates are the transmission zeros of the equilibrium price response, and none cancels (Theorem 5.3(i) with the effective parameters).*

(iii) Boundary. *The effective merged measure is $\tilde{M} = h\, M$, where $h(r) = (D - \bar{a})/(D - r)$ is positive. The effective parameters determine the equilibrium ones: $D = 1 - \kappa^* \tilde{M}([0, 1])/(1 - a^*)$, $\bar{a} = a^* D$ and $\kappa = \kappa^* D$. So the merged measure remains exactly what equilibrium prices identify (Theorem 3.3).*

(iv) The rates survive. *Let $(a^*, \kappa^*, W, r_c, \Delta\eta)$ satisfy the hypotheses of Theorem 5.1 with $(a, \kappa) = (a^*, \kappa^*)$. Then the two K-cohort economies of that theorem are the price responses of two inventory equilibria with identical primitives $(A, \sigma_F^2, \sigma_\xi^2, \bar{a})$ and identical (κ, D):*

- *$D = 1 - \kappa^* W/(1 - a^*)$, which exceeds every rate;*
- *$\bar{a} = a^* D$ and $\kappa = \kappa^* D$;*
- *any $\sigma_F^2, \sigma_\xi^2 > 0$, with $A = \kappa^* D(1 - a^*)/(\sigma_F^2 + \kappa^{*2} \sigma_\xi^2)$;*
- *cohort weights $w_i = \tilde{w}_i (D - r_i)/(D - \bar{a}) > 0$.*

Hence an observer of equilibrium prices who knows the primitives needs $N \geq c\, \sigma^2 (\Delta\eta)^{-(4K-2)}$ experiments. Theorem 5.2 applies to the effective parameters, and the true weights follow from the effective ones by a Lipschitz map. So $N^* \asymp \sigma^2 (\Delta\eta)^{-(4K-2)}$ in equilibrium.

In words. Dealers' anticipation does not hide the cohorts and does not move them. It replaces the persistence and impact by $\bar{a}/D$ and κ/D, and it rescales each cohort's weight by $(D-\bar{a})/(D-r_i)$. Because that factor rises with r_i, slow-memory cohorts weigh more in equilibrium prices. The memory rates are untouched.

- Every result of Sections 3 and 5 that concerns the map from order flow to prices therefore holds in equilibrium: identification, sample complexity, passive observation, and probing costs computed from the price path. Its constants are evaluated at the effective parameters.
- The lower-bound pair is not an artefact of the reduced form. Two economies with the same dealers, the same risk aversion and the same impact differ only in their cohorts, and they are as hard to tell apart as Theorem 5.1 says.
- Part (i), the realization in part (iv) and the effective cohort recursion are machine-checked (`Paper1EquilibriumReducedForm.lean`).
- One assumption is new: the observer's probe is part of the exogenous flow that dealers price as noise. Dealers who recognized a probe would price it differently.

Further equilibrium results. Four results in Supplement S2 extend this section. None is needed for the main argument.

- *A strategic insider* with short-lived information (Theorem S2.3). The equilibrium is characterized in closed form and nests Kyle's one-period values. Its coefficients depend on the cohorts only through G, and the observed price is the closed loop driven through a known invertible moving-average filter. So the identification boundary and the $(\Delta\eta)^{-(4K-2)}$ rate hold with an insider present. Positive feedback raises total impact, and the insider then trades less and prices reveal less of her news.
- *Fragility* (Theorem S2.2). With positive feedback, a positive-impact equilibrium exists only if dealer risk aversion lies below a threshold. For strong feedback the threshold is inversely proportional to the feedback intensity $G(1)$. With several cohorts the bound depends on the population only through $G(1)$, again an aggregate.
- *Rents* (Theorem S2.4). A dealer that misjudges the aggregate response by ε hands an insider with a committed schedule a rent of exact order ε, or charges it one. Public data reveal the aggregate response at the ordinary rate. So the information that protects price formation is cheap to learn.
- *A targeted halt in equilibrium* (Theorem S2.5). Two markets that agree in equilibrium to within the cube of the memory spacing can differ at order one after their largest

group is halted.

Disclosure keeps its value in equilibrium too (Corollary S2.1).

5 The price of composition

Prices identify the kernel, and with exponential memory the kernel determines the merged measure. A finite record, however, pins down only finitely many moments of M, each with noise. This section asks how much data it takes to recover the groups. The answer is the known rate for finite mixtures. What the market adds is that the memories must first be extracted from a feedback loop, and that waiting is no cheaper than trading.

5.1 Many unknown cohorts

Theorem 5.1 (K-cohort two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0,1)$, total merged weight $W > 0$ and $0 < \Delta\eta < \min(r_c, 1 - r_c)$, so that all rates lie in $(0,1)$, and assume $a + \kappa W/(1 - r_c - \Delta\eta) < 1$, so that every candidate economy is Schur stable. Each reset experiment observes the noisy taps $h_0, \dots, h_{T-1}$ with $T \geq 2K + 1$ ($T = \infty$ allowed), so that the first tap where the two economies below differ, h_{2K}, is observed; the bound below uses the full response norm, which dominates any finite prefix. There exist two K-cohort economies with rates in $r_c + \Delta\eta[-1, 1]$, positive weights, and merged measures whose moments of order $0, \dots, 2K-2$ coincide while the moment of order $2K-1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from N independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.

The two losses have different targets. The weights of the two economies differ by a constant that does not shrink with $\Delta\eta$, so the first statement is about weights at fixed accuracy, which Theorem 5.2 matches; the Wasserstein statement is about the memory distribution at accuracy proportional to $\Delta\eta$, which the full working paper matches with an upper bound.

The construction uses the K-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have K atoms and integrate polynomials of degree $2K - 2$

exactly. For $K = 2$ it reproduces the order of Theorem S4.1. Numerically, their low moments agree to machine precision and their moment of order $2K - 1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with K on top of the exponent. The two rules also differ in their largest weight for every K we checked: for $K = 2, \dots, 200$ the largest Gauss–Radau weight exceeds the largest Gauss weight, by 0.25 at $K = 2$ and by about $0.79/K^2$ for large K (`equilibrium/gauss_radau_maxweight.py`). So the argument given for $K = 2$ in Proposition 6.1 applies to each such K. A proof for every K is open.

What counts as an experiment. In Theorem 5.1 an experiment is a unit impulse from the reset state whose response is observed with noise of variance σ^2 per tap. The exponent is the same for any probe design of bounded energy. For any probe v the mean difference between the two economies is at most $\sup_\omega |\Delta\hat{H}(e^{i\omega})|\,\|v\|_2$, and $\sup_\omega |\Delta\hat{H}| = O((\Delta\eta)^{2K-1})$ because $\Delta\hat{H} = z^{-1}\hat{H}_A\hat{H}_B\,\Delta G$ with both responses bounded on the circle. Experiments with probe energy E each therefore need $NE \gtrsim \sigma^2(\Delta\eta)^{-(4K-2)}$: a larger probe buys the same information with fewer experiments, and the account-level analysis of the full working paper prices the capital that a larger probe requires. The bound does not cover information obtained without trading against the groups, such as class records (Section 6).

Machine check. Lean proves the bound for every K and every horizon T, with a constant that does not depend on T. It uses the theorem's stability hypothesis $|a| + |\kappa|W/(1 - r_c - \Delta\eta) \le \varsigma < 1$, and holds for all error probabilities with $\alpha_0 + \beta_0 \le 1$ (`p1_thm_kcohort_lower_bound_uniform`): $4\sigma^2(1-\alpha_0-\beta_0)^2 \le N\,C_U^2(\Delta\eta)^{4K-2}$. The checked pair is a different moment-matched pair with the same order: binomial weights on $2K$ equally spaced nodes, split into odd and even halves. For the Gauss and Gauss–Radau rules themselves, Lean checks the case $K = 3$ (positive weights, matched moments and the fifth-moment gap). Lean also checks both estimator statements for this pair, for every horizon (`p1_thm_kcohort_lower_bound_weights`, `p1_thm_kcohort_lower_bound_W1`). The first covers an estimator of the weights, listed in order of memory rate, that is accurate to $LW/4^L$ in each coordinate with probability at least 3/4 under both economies. The second covers an estimator of the merged measure that integrates every 1-Lipschitz function vanishing at zero to within $2\Delta\eta W/(\pi(2K-1))$. For measures of equal mass, a Wasserstein error below that accuracy implies this. Both force $\sigma^2 \le N\,C_U^2(\Delta\eta)^{4K-2}$, here with $L = K - 1$. The matching lower order $\|h^A - h^B\|_2 \gtrsim (\Delta\eta)^{2K-1}$ is proved on paper.

Theorem 5.2 (Matching upper bound for every K)**.** *Fix $K \ge 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i\ne j}|s_i - s_j| \ge \gamma > 0$, and merged*

weights $w_i \in [\omega_{\min}, \omega_{\max}]$ *with* $\omega_{\min} > 0$. *Consider the class of* K*-cohort economies with rates* $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, *and* N *independent reset experiments, each observing at least the taps* $h_0, \dots, h_{2K}$ *with Gaussian noise of variance* σ^2. *There are constants* $c_0, C > 0$ *depending only on* $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ *such that, whenever* $\epsilon_N \equiv \sigma N^{-1/2} (\Delta\eta)^{-(2K-1)} \le c_0/\sqrt{\log(4K/\beta)}$, *Prony's method applied to the first* $2K$ *estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \;\le\; C\sqrt{\log(4K/\beta)}\;\epsilon_N$$

with probability at least $1 - \beta$. *Hence* $N = C'\sigma^2 (\Delta\eta)^{-(4K-2)}$ *experiments suffice for any fixed accuracy and confidence. On any such class that contains the Gauss and Gauss–Radau pair of Theorem 5.1 (a rich class), together with that theorem the minimax sample requirement is* $N^* \asymp \sigma^2 (\Delta\eta)^{-(4K-2)}$. *On smaller classes the upper bound need not be tight: if the rates are known, for example, two weights need only order* $\sigma^2 (\Delta\eta)^{-2}$ *experiments. For* K *known clustered rates the requirement is at least of order* $\sigma^2 (\Delta\eta)^{-2(K-1)}$. *A weight perturbation along the* $(K-1)$*-th finite difference preserves the moments of order* $0, \dots, K-2$ *and changes the moment of order* $K-1$ *by order* $(\Delta\eta)^{K-1}$. *For* $K = 3$, *the direction* $(1, -2, 1)$ *on rates* $r_c - \Delta\eta, r_c, r_c + \Delta\eta$ *changes the second moment by* $2(\Delta\eta)^2$. *This is still far below the unknown-rate order* $(\Delta\eta)^{-(4K-2)}$. *The constants* c_0 *and* C *grow quickly with* K, *through the conditioning of the moment map; the theorem gives the order in* $\Delta\eta$ *for fixed* K, *not a usable constant for large* K *(in the full working paper our double-precision Prony implementation already fails at* $K = 4$ *on the disclosure grid).*

In words. With K unknown groups whose memories lie within $\Delta\eta$ of each other, the number of experiments needed grows like $(\Delta\eta)^{-(4K-2)}$. For two groups, halving the spread multiplies the data needed by 64; for three groups, by 1,024. Theorem 5.1 shows that no method does better, and this theorem shows that Prony's method achieves it.

Machine check. Lean verifies this theorem for every $K \ge 1$, using only Lean's standard axioms. The formal proof follows the one above: a Lipschitz bound for the closed-loop deconvolution that does not depend on $\Delta\eta$, the $(\Delta\eta)^{-m}$ rescaling, a sub-Gaussian tail bound over the $2K$ taps, injectivity of the Jacobian of the moment map (proved by Hermite interpolation, without the determinant formula), and a uniform inverse function theorem on the compact class. The formal estimator is the representation with increasing nodes and nonzero weights that matches the $2K$ estimated moments; Lean shows that it is unique and that Prony's linear system has a unique solution whose polynomial has exactly its nodes as

roots, so the roots are real. The formal statement allows sub-Gaussian noise and assumes $|r_c| + \Delta\eta \leq 1$, so that every memory rate has modulus at most one.

5.2 Passive data

Theorem 5.3 (Passive identification: minimum phase)**.** *Consider a stable economy with $0 < a < 1$, distinct memory rates $0 < r_i < 1$ and nonzero merged weights.*

(i) The zeros of H are the memory rates r_i together with a structural zero at $z = 0$; none cancels against the denominator, and all lie in the open unit disk, so H is minimum phase. The degenerate anchor ($a = 0$) and zero-rate cases require the cancellations described in the proof of Theorem 3.3.

(ii) Let the price be driven by a single innovation sequence with variance $\sigma_\xi^2 > 0$, possibly unknown, so that its spectrum is $f(\omega) = \sigma_\xi^2 |H(e^{i\omega})|^2/(2\pi)$. Within the class of (i), f determines $\sigma_\xi^2\kappa^2$ and the normalized response H/h_0 (recall $h_0 = \kappa$); hence it determines the memory rates. If in addition κ (or σ_ξ^2) and a are known, it determines H and, by Theorem 3.3, the merged measure.

Machine check. For every K, Lean checks part (i) and part (ii) starting from equal spectra at every frequency (`p1_thm_minphase_zeros`, `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged`). The check uses H in the rational form $\kappa z D(z)/((z - a)D(z) - \kappa N(z))$ of Theorem 3.3, and *stable* in the sense that the poles of H lie in the open unit disk. Both are derived from the market recursion. The eigenvalues of the transition of the state $(x_t, y_{1,t}, \ldots, y_{K,t})$ are exactly the zeros of the denominator of H, so Schur stability of the recursion is the pole condition (`p1_eigen_iff_bottom_root`, `p1_schur_stable_iff_poles_in_disk`). The generating function of the recursion's price response h_t to a unit trade satisfies $\left(\sum_t h_t x^t\right) x^{K+1} B(1/x) = x^{K+1}\kappa x^{-1} D(1/x)$ as formal power series, where B is the denominator of H; with $x = 1/z$ this is $\sum_t h_t z^{-t} = H(z)$ (`p1_impulse_generating`, `p1_tfBottomRev_eval`, `p1_tfTopRev_eval`).

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the nonstructural zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the cohorts' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

Theorem 5.4 (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of i.i.d. Gaussian innovations $\xi_t \sim N(0, \sigma^2)$, $\sigma^2 > 0$ known, without observation noise, for θ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

(i) For a single cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.

(ii) For the moment-matched two-cohort economies of Theorem S4.1, $D = \Theta(\tau^2(\Delta\eta)^6)$.

(iii) For every $K \geq 2$, for the K-cohort Gauss and Gauss–Radau economies of Theorem 5.1, $D = \Theta((\Delta\eta)^{4K-2})$.

Hence any test based on T passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, uniformly as the separation shrinks: passive data cost at least the same polynomial order as active probing for every number of cohorts. For each fixed pair, at fixed type-I error the likelihood-ratio test's type-II error decays at exponential rate D (Stein's lemma), and both errors decay at the Chernoff-information rate, which is of the same order in the separation; so the orders are the right ones; we do not claim that this attainment is uniform as the separation shrinks. All constants depend only on K, the stability margin of the family, the rate interval $[\eta_{\min}, \eta_{\max}]$, the total weight and the moment-matched construction; they do not depend on δ, $\Delta\eta$ or T. In part (ii) τ is held fixed as $\Delta\eta \to 0$. The testing conclusions use a finite-record bound $D_{\mathrm{KL}}(P_1^T \| P_0^T) \leq C\,T\,\varepsilon^2$, with ε the leading moment gap (δ^2, $\tau(\Delta\eta)^3$, $(\Delta\eta)^{2K-1}$); the lower bounds on D and the fixed-pair error exponents are separate statements.

Machine check. Lean checks the finite-sample consequence of parts (i) and (iii), not the divergence-rate statements. In the checked record the market starts from rest and runs for any number M of warm-up periods before the T recorded prices. The bound does not depend on M, but the passage to the stationary law ($M \to \infty$) is not checked. Part (iii) is checked for every pair of populations with nonnegative merged weights of total at most W, normalized rates in $[-1, 1]$ and equal moments below order $2K - 1$ (`p1_thm_passive_cost_matched`). This covers the K-cohort Gauss and Gauss–Radau economies of Theorem 5.1: both rules are positive with nodes in $[-1, 1]$ and integrate polynomials of degree $2K - 2$ exactly. Lean proves these properties of the two rules for $K = 2$ and $K = 3$ (`p1_thm_passive_cost_gauss_radau_two`, `p1_thm_passive_cost_gauss_radau_three`), so for these K the check covers the paper's economies with no extra hypothesis. For general K, the exactness of the two quadrature

rules, a classical property, enters the check as the moment hypothesis. It also covers the binomial pair used in the numerical checks (`p1_thm_passive_cost_every_K`).

In words. Watching the market without trading does not escape the barrier. A passive observer needs at least the same polynomial order of observations as a prober needs experiments; it pays in time rather than in trading losses. Prices recorded with extra measurement noise that is independent of the price are a garbling of the noiseless record, so by the data-processing inequality they carry less information, and the lower bound holds a fortiori.

In executed quantity and money. For a *known* pair (one group at rate η against two at $\eta \pm \delta$ with the same total weight), the information needed can be stated in trades. The executed quantity a prober needs is of exact order δ^{-2}: the two price responses first differ at lag two, by $-\kappa^2 q \delta^2$, and a single pulse of size $2\sigma\Phi^{-1}(1-\alpha)/(\kappa^2 q\delta^2)$ attains error α. The capital a counterparty must post to fill such probes is of order δ^{-4}, because the adverse mark of a probe grows with the square of its size. Halving the difference between two memories therefore quadruples the trading needed and multiplies the capital by sixteen (Theorems S3.1 and S3.3 and Lemma S3.5, Supplement S3).

6 Who needs composition, and what disclosure buys

By Theorem 3.1, composition is worth nothing to anyone trading on public information. It can still matter to a supervisor whose action is aimed at one group. This section gives a decision in which it does, and the disclosure that removes the cost of learning it.

A decision in which attribution matters. A supervisor must keep the feedback budget $S = a + \kappa \sum_i q_i$ below a prudential ceiling $\bar{s}$. It can either cut all activity uniformly or halt the largest verified class. The target is the feedback budget, not the spectral radius of the state matrix: in an already stable positive economy, imposing $S \le \bar{s} < 1$ is a tighter prudential constraint, not a restoration of stability.

Proposition 6.1 (Supervisory separation: market-wide versus targeted intervention)**.** *Assume the positive stable model, known a, κ, and $a < \bar{s} < 1$. Write $q_i = w_i/(1-r_i)$ and $R = S - \kappa \max_i q_i$ for the post-halt feedback budget.*

(i) A uniform scale θ meets the ceiling exactly when $a + \theta(S-a) \le \bar{s}$. Its largest feasible value is $\theta^* = \min\{1, (\bar{s}-a)/(S-a)\}$. The identity $S = 1 - \kappa/\sum_j h_j$ is algebraic; estimating this infinite sum requires a justified tail bound and an observation/noise model. It alone establishes no unconditional parametric learning rate.

(ii) For $K = 2$, the Gauss and Gauss–Radau pair of Theorem 5.1 has limiting largest normalized weights $1/2$ and $3/4$. Thus $|S^G - S^R| = O((\Delta\eta)^3)$ and $q^R_{\max} - q^G_{\max} = W/(4(1 - r_c)) + O(\Delta\eta)$. For sufficiently small spacing, a nonempty interval of ceilings separates their post-halt decisions. For N independent reset records with common Gaussian noise $\sigma^2 I$, both decision errors at most $1/4$ require

$$N \geq (2\Phi^{-1}(0.75)\sigma/\|h^G - h^R\|_2)^2 \geq c\sigma^2(\Delta\eta)^{-6}.$$

The record horizon and the response norm must be the same on both sides.

(iii) Tagged-flow estimation under Theorem 6.3, a rate bound away from one, and a fixed nonzero decision margin gives a bounded sample requirement as the spacing shrinks. The target class must be mapped to an executable, verified class label; a largest latent atom alone supplies no such identity.

The proof and the general-K scope are in Supplement S5.

In words. A uniform cut needs only the kernel, through $S = 1 - \kappa/\sum_j h_j$, so public prices suffice for it. A halt of the largest group needs that group's weight, which the kernel does not pin down. The two-group populations of Theorem 5.1 (the Gauss and Gauss–Radau pair with $K = 2$) have feedback budgets within $O((\Delta\eta)^3)$ of each other, but largest groups that differ at order one. So a range of ceilings exists at which halting the largest group meets the ceiling in one population and not in the other. Prices decide between them only with order $(\Delta\eta)^{-6}$ records.

6.1 What class-tagged disclosure buys

Proposition 6.2 (Tagged flow removes the barrier)**.** *Let the price x_t be the causal stationary solution of the model driven by i.i.d. centred innovations of variance $\sigma_\xi^2 \in (0, \infty)$ in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose cohort i has merged weight $w_i > 0$ and the observer receives prices and each cohort's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. centred measurement noise of finite variance σ_f^2, independent of the whole innovation sequence and hence of the whole price path. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2 \big[(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)\big], \quad (11)$$

where $A_m = \sum_{s\geq 0} r_i^s \gamma_x(s+m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ *and* $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. *The Jacobian determinant is*

$$\det \Sigma_i = w_i \sum_{\ell,m} c_\ell c_m \frac{1-\lambda_\ell \lambda_m}{1-r_i \lambda_m} \;>\; 0. \tag{12}$$

V_i *depends only on cohort* i*'s own* (w_i, r_i), σ_f^2 *and the price autocovariances. Two conclusions are distinct: (a) for each admissible economy,* $\hat{\beta}_i$ *has the stated limiting normal law; (b) on a uniformly stable family with stationary price variance at least* $\gamma_{\min} > 0$ *and merged weights at least* $w_{\min} > 0$*, the limiting covariance* V_i *is bounded by a constant that does not depend on* $|r_i - r_j|$*. Statement (b) bounds the limit; it is not a central limit theorem that holds uniformly over the family. Three pieces are machine-checked: the finite-sample reset-experiment version, with an error bound that does not depend on the spacing; the lower bound on the determinant of the population instrument matrix that does not depend on the spacing (**p1_tagged_iv_determinant_bound**); and the exact decomposition of the IV score into reporting-error terms with coefficients fixed by the price path plus two boundary terms (**p1_iv_boundary_decomposition**). The stationary law of large numbers and the central limit theorem behind (a) are not machine-checked.*

In words. If each group's trading is reported separately, each group becomes an ordinary regression. Its precision depends on the group's own rate and on the price dynamics, but it has no singularity as the groups' rates approach each other.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about w_i does not involve the other cohorts' rates; the closed form (12) matches the simulated Jacobian (0.1330 against 0.1322 in a two-cohort economy with 4×10^5 periods). In simulation (T = 5,000, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between 5.8×10^{-4} and 6.6×10^{-4}). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem. Proposition 6.2 says that tagging removes the barrier; the next result says by how much.

Theorem 6.3 (Value of class-tagged disclosure)**.** *Take a rich class of Theorem 5.2: rates* $r_c + \Delta\eta\, s$ *with* s *in a compact set of* γ*-separated ordered points that has nonempty interior and contains the Gauss and Gauss–Radau pair of Theorem 5.1, and merged weights in a nondegenerate interval* $[\omega_{\min}, \omega_{\max}]$*,* $0 < \omega_{\min} < \omega_{\max}$*. In each reset experiment the observer applies a unit impulse and records the price response with independent* $N(0,\sigma^2)$ *noise per tap; under tagging it also records each class's flow response* $f_{i,t} = w_i \sum_{s\leq t} r_i^{t-s} h_s$ *with independent* $N(0,\sigma_f^2)$ *noise per tap. Let* N_p *and* N_t *be the numbers of experiments needed, without and with tagging, for a fixed accuracy and confidence. There are constants* $0 < c \leq C < \infty$*,*

depending on (σ, σ_f) *and the class but not on* $\Delta\eta$*, such that*

(i) to recover the weights to a fixed accuracy smaller than half the weight difference between the Gauss and Gauss–Radau economies (so that recovering the weights decides between them), $c \le N_t \le C$*, so* $N_p/N_t \asymp (\Delta\eta)^{-(4K-2)}$*;*

(ii) to recover the rates to accuracy $c_0\Delta\eta$*, for a fixed* $c_0 > 0$ *smaller than the radius of a ball contained in the class of rescaled nodes (so that no fixed guess meets the target),* $c(\Delta\eta)^{-2} \le N_t \le C(\Delta\eta)^{-2}$*, so* $N_p/N_t \asymp (\Delta\eta)^{-(4K-4)}$*.*

In words. This is the value of disclosure measured in data. Tagging replaces a requirement that explodes as the groups become similar with one that does not grow at all (for the weights) or grows only like $(\Delta\eta)^{-2}$ (for the rates).

Proposition 6.2 and Theorem 6.3 use different observation arrangements: the proposition concerns passive, stationary data with prices observed without noise, where the price is a valid instrument; the theorem concerns reset experiments with noisy prices. With noisy prices, regressing flow on the observed price is an errors-in-variables problem, which is why the theorem works from the known impulse instead.

7 Scope and limits

Linearity. Sufficiency (Theorem 3.1, Proposition 3.2) needs aggregate feedback flow to be a linear function of past public signals, plus independent idiosyncratic shocks if there are any. With cohort-specific Gaussian demand shocks, a second aggregate enters, the spectral density of the aggregate shock process, and the equivalence class shrinks accordingly (Supplement S2). Nonlinear rules, such as threshold strategies, are outside the theorem. The full working paper reports simulations of stylized nonlinear rules and of online learners trained in the closed loop. In those simulations, the costs of Section 5 hold when the competing populations' feedback weights match. Populations whose learned weights differ are told apart by those weights instead. Hard thresholds hide groups from small probes altogether. These are simulations, not theorems.

Exponential memory. The identification and cost results need memory to be a finite mixture of exponentials. For general completely monotone memory, the full working paper proves that the minimax accuracy for the memory distribution is of exact order $1/\log(N/\sigma^2)$. Halving the error then squares the data needed, instead of multiplying it by a constant.

The observer. The lower bounds hold for an observer who knows (a, κ) and the noise law, so they hold a fortiori without that knowledge. The upper bounds assume reset experiments or a known initial state. Strategic responses to being observed are not modelled: the cohorts do not change their rules when probed.

Evidence. This version reports no empirical test. The full working paper reports pre-specified tests on two sets of trader-class reports: the CFTC's weekly Commitments of Traders in twenty futures markets and Binance's class ratios in 30 perpetual markets. Of 23 recorded outcomes, 11 pass. The class reports constrain the persistence of class positions far more tightly than prices do. They do not identify learning memory, because that persistence can also reflect turnover, reporting aggregation or inventory retention. A pre-registered out-of-sample test of the feedback premise on Taiwan's class-tagged exchange flows is under way. Its holdout period has not been opened, and no result from it is reported here.

Open problems. Three questions remain open: a long-lived strategic insider who times her trades against the cohorts' memory; sufficiency and its failure for nonlinear populations; and a supervisory loss function estimated from real interventions, which would turn the information costs here into a value of disclosure.

8 Conclusion

In a linear market with feedback traders, prices summarize the trader population in one object, the feedback kernel. The kernel is sufficient for every outcome that price-based trading determines, for any insider and any causal dealer rule. It is all that prices identify, and with inventory-averse dealers both statements hold in equilibrium with unchanged memories. Information beyond the kernel, who the groups are and how much each trades, is worth nothing to anyone trading on public information. It is worth something to a supervisor whose action is aimed at a particular group. Learning it from prices costs of order $(\Delta\eta)^{-(4K-2)}$ experiments or observations, executed quantity of order δ^{-2} and capital of order δ^{-4}. Class-tagged flow removes that cost.

The practical reading is a division of supervisory labour. Market-wide measures, such as limits on aggregate feedback intensity or confidence sets for the price response, can be built from public data, because they depend on the kernel alone. Targeted measures cannot be built from public data at reasonable cost when the groups' memories are close. For targeted measures, verified class-level records are the efficient source of information, not more price data or more sophisticated estimation.

Declarations

Funding. This research received no external funding.

Competing interests. The authors declare no competing interests.

Author contributions. Pratik Niroula: conceptualization, methodology, software, formal analysis, writing (original draft). Proposed review responsibilities for confirmation: Prashant Niroula and Sujan Khatiwada, proof review, validation and manuscript review. Independent human proof review and final approval by each author remain to be recorded; this draft does not attest that they have occurred.

Data availability. This version uses no market data. The data used by the full working paper are public and are listed in its data statement.

Code availability. All code, pre-specified test specifications, result files and test-unlock counters are public at `https://github.com/Pratikn03/Fianince`. The repository's README lists the scripts in run order, and the full working paper's reproducibility appendix maps every number to the file that produces it.

Ethics approval. Not applicable. The study uses only public market data and involves no human participants or personal data.

Use of AI tools. Claude (Anthropic) assisted with data analysis, analysis code, derivations, numerical results and text. Gemini (Google) assisted with planning and internal draft critique. ChatGPT/Codex (OpenAI) assisted with planning, literature discovery, mathematical development, code and Lean proof drafting, executable and kernel checks, result interpretation and manuscript revisions. This assistance was material. The reported kernel checks and computational runs support their explicitly stated results; they do not certify independent human review, novelty or full-manuscript correctness. AI tools are not authors. The human authors remain responsible for verifying all content, citations, analyses and conclusions and for the final scholarly contribution before submission.

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